Real-Time All-Digital Carrier and Timing Synchronization Unit

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01 Project Introduction

- The goal of this project is to design, implement, and verify a real-time, hardware-efficient, all-digital baseband synchronization subsystem on an FPGA/ASIC platform. Targeted for burst-mode wireless receivers (e.g., 5G NR, Wi-Fi 6/7, satellite communications), the subsystem accepts oversampled baseband I/Q samples, corrects symbol timing errors, estimates coarse carrier frequency offset (CFO), and performs joint fine CFO and phase tracking for QPSK and 16-QAM constellations.
- Teams must optimize the pipeline for real-time throughput, streaming fixed-point operation, low latency, minimum logic utilization, and low power while maintaining structural modularity and robust testability.

02 Architecture Specifications



Figure 1 Top-level Digital subsystem Interface block diagram

2.1. Battery and operating conditions

ReqNr	Symbol	Description	Min	Typ	Max	Unit	Remark
R.101	F_{clk}	System Clock Frequency	100	125	200	MHz	Primary FPGA Clock
R.102	N_s	Oversampling Factor	—	4	—	Sps	4 samples per symbol
R.103	R_{sym}	Symbol Rate	25	31.25	50	MSps	Derived from F_{clk} / N_s
R.104	W_{in}	Input Data Bit-width	—	12	—	Bits	Signed Q1.11 per channel
R.105	W_{out}	Output Data Bit-width	—	16	—	Bits	Signed Q1.15 per channel
R.106	T_{pip}	Steady-State Pipeline Latency	—	—	64	Cycles	After lock at target F_{clk} , excludes acquisition
R.107	Modulation	Target Constellations	QPSK, 16-QAM				Runtime selectable

2.2. Symbol Timing Recovery (STR)

ReqNr	Symbol	Description	Min	Typ	Max	Unit	Remark
R.108	TED_type	Timing Error Detector	Gardner TED			—	Non-data-aided
R.109	Interp	Interpolator Structure	Farrow (Parabolic/Cubic)			—	Fractional delay
R.110	W_{mu}	Fractional Delay (μ) Precision	8	10	12	Bits	Unsigned fixed-point

R.111	e_{str_jitter}	Residual Timing Jitter (RMS)	—	—	0.02	Ts	At SNR = 15 dB
R.112	T_{str_lock}	STR Lock Acquisition Time	—	—	150	Symbols	To steady-state delay

2.3 Coarse & Fine Carrier Frequency Offset (CFO) Recovery

ReqNr	Symbol	Description	Min	Typ	Max	Unit	Remark
R.113	$CFO_{capture_Coarse}$	Coarse CFO Capture Range	-0.10	—	+0.10	R_{sym}	±10% of symbol rate
R.114	CFO_{pullin_fine}	Fine CFO pull-in Range	-1	—	+1	R_{sym}	at least 2× the coarse residual to guarantee lock
R.115	$Algo_{coarse}$	Coarse Estimation Block	FFT / Conjugate Auto-Corr			—	Preamble-assisted
R.116	T_{coarse}	Coarse CFO estimation latency	32	64	128	Symbols	Preamble-assisted
R.117	CFO_{res}	Steady-state residual CFO			0.01	R_{sym}	After fine lock
R.118	Loop_Type	Fine Carrier Tracking Architecture	Decision-Directed Costas Loop			—	PI Loop Filter
R.119	CFO_{res_fine}	Fine CFO resolution (steady-state)			0.01	R_{sym}	After lock; determined by NCO LSB and loop BW
R.120	W_{nco}	NCO Accumulator Precision	32	32	32	Bits	Phase resolution

R.121	φ_{res}	Residual Phase Error (RMS)	—	—	1.5	Deg	At SNR ≥ 15 dB
R.122	T_{costas_lock}	Fine Lock Acquisition Time	—	—	200	Symbols	Continuous tracking
R.123	ζ	Costas loop damping factor	0.5	0.707	1.0	—	0.707 = critically damped
R.124	B_n	Costas loop noise bandwidth	0.02	0.03	0.05	Rsym	Normalized to symbol rate

03 Checkpoints & Marking Scheme

Checkpoint 1: : Architecture, Fixed-Point Modeling & Budgeting (15 Points)

• Develop a golden bit-accurate MATLAB/Python fixed-point model of the complete unit.
• Document bit-width allocation (word and fraction lengths), overflow management, and truncation strategies.
• Calculate loop filter dynamic parameters (ω_n , ζ) to meet lock time and stability criteria.

Checkpoint 2: Symbol Timing Recovery (STR) Module Design (15 Points)

• Implement synthesizable RTL for Farrow Interpolator, Gardner TED, and proportional-integral loop filter.
• Verify correct tracking and strobe output generation under fractional delay offsets ($0.0 \le \tau < 1.0$).

Checkpoint 3: Coarse CFO Estimation Module Design (15 Points)

• Implement hardware preamble-assisted coarse frequency detector (FFT or auto-correlation).
• Demonstrate estimation accuracy within ±0.5% of Rsym over SNR range from 0 dB to 20 dB.

Checkpoint 4: Fine CFO & Costas Phase Tracking Loop (15 Points)

- Implement Decision-Directed Costas Loop, 32-bit NCO, and CORDIC/Complex multiplier derotator.
- Verify phase acquisition and constellation de-rotation for both QPSK and 16-QAM schemes.

Checkpoint 5: Design for Testability (DFT) Infrastructure (10 Points)

- Integrate 16-channel observation MUX to output internal status signals (TED error, NCO phase) to test bus.
- Provide force registers allowing manual override of loop control registers for individual block isolation.

Checkpoint 6: Top-Level Integration & Performance Verification (20 Points)

- Integrate the full receiver pipeline and execute burst frame simulations with AWGN and channel skews.
- Achieve RMS EVM ≤ 2.5% in clean conditions, and RMS EVM ≤ 5.5% (QPSK) / ≤ 8.0% (16-QAM) at SNR ≥ 15 dB, including residual CFO and timing offsets.
- EVM should be measured at the output of the synchronization subsystem after derotation and timing correction with root-raised cosine (RRC) pulse shaping with a roll-off factor (α) of 0.35; over at least 1000 symbols.

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Checkpoint 7: Post-Synthesis & Post-Implementation Verification (10 Points)

- Verify that the RTL design, after synthesis and implementation, still meets all functional requirements.
- Achieve timing closure at ≥ 125 MHz and provide resource utilization (DSP48, BRAM, LUT) and power consumption analysis.

04 Bonus Criteria

EXTRA BONUS

4.1 General Report & RTL Quality Guidelines (10 Points)

- Submit professional documentation with clear hierarchical state-machine diagrams and cycle-accurate waveform figures.
- Enforce clean coding standards: parameterized Verilog modules, robust reset synchronizers, and structured handshake signals.

4.2 Gate-Level Power Analysis (5 Points)

- Accurate dynamic and leakage power characterization using switching activity input files (.saif or .vcd) derived from post-synthesis gate-level simulations.

4.3 Download the design into FPGA board (10 Points Bonus)

- Download your design on an FPGA (e.g., Intel, or AMD/Xilinx Artix-7) and verify functional and timing requirements.

4.4 ASIC Implementation & Synthesis Optimization (10 Points Bonus)

- Synthesis Flow: Standard-cell mapping using industry-standard tools (e.g., Synopsys Design Compiler or Cadence Genus).
- Metrics & Reports: Total area, gate-level timing closure (worst negative slack/setup & hold), and cell count.